ANAND RAMNARAIN

Graphics & Engine Programmer — C++ · DirectX 12 · HLSL

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Graphics and engine programmer completing a Master's in Advanced Programming for AAA Games at UPC Barcelona. I've built Phoenix, a DirectX 12 engine expanded on my own from coursework, and Arclight, a C++20 engine written from scratch for DirectX 12 and Metal. I enjoy finding the root cause of problems and working with artists and designers.

PERSONAL PROJECTS (source code on GitHub)

Phoenix Engine — *C++ / DirectX 12, grew out of Master's coursework* Jan 2026 – present · source

- **Independent development:** expanded the graded coursework model viewer (v0.2) into a full engine, building every system below myself. Now used to develop the game Ashfall Keep.
- **Rendering:** realistic materials lit by many lights at once (deferred PBR, image-based lighting, compute tile-based light culling), plus decals, water, transparency and particles.
- **Shadows and GPU compute:** cascaded sun, spot and point-light shadows (PCF, VSM, ESM); skinning, morph targets, particles, light culling and volumetric fog as compute shaders; bloom and post-processing.
- **Engine systems:** templated ECS-style components, JSON scenes and prefabs, hot-reloading game scripting, physics with swappable broad phases, and A* pathfinding with steering.
- **Debugging and optimisation:** fixed GPU resource-lifetime and window-resize crashes and upside-down animated models; removed stutters by caching descriptor tables.

Arclight Engine — *C++20 engine and editor, written from scratch for Windows and macOS* v0.3.0 · source

- **Cross-platform rendering:** my own rendering layer (RHI) running DirectX 12 on Windows and Metal on macOS; shaders written once in HLSL, converted at build time, every feature tested on both.
- **GPU memory safety:** leak-reporting resource registry, generational handles, fence-tagged deferred release and a per-frame upload ring buffer.
- **Tooling and tests:** own maths library, SDL3, CMake, ImGui editor, glTF 2.0, GPU capture (RenderDoc, PIX, Xcode), and GitHub Actions running ~78 unit tests on both platforms.

Cross-Engine Spatial Data Streaming SDK — *C++20 library* 2026 · source

- Streams large 3D map datasets within a fixed memory budget (quadtree, level of detail, background loading); one interface for Unity, Unreal and DirectX 12, with 187 automated tests.

TEAM GAME PROJECT

Bound by Death — *co-op hack-and-slash, Ravenwhisp Studio, UPC Master's* 17-person team · site

- **Rendering and VFX:** light culling (checked on the GPU with PIX) and post-processing in the team's DirectX 12 engine; character VFX, a low-health heartbeat pulse and a death fade-and-blur.
- **Engine tools and teamwork:** built the prefab system and a controller-vibration system designers tune without code; Scrum sprints, code reviews, and close work with artists and designers.

EXPERIENCE

Software Developer (vacation work) — *BBD Software* Nov 2024 – Dec 2024

- Built a team web application with React/TypeScript and PostgreSQL, deployed with Docker on Azure.

Web Developer (freelance) — *Easy Shoe & Bag Repair* Jan 2023 – Nov 2023

- Built and maintained a client website, product catalogue and SEO.

EDUCATION

Master's in Advanced Programming for AAA Games — *UPC, Barcelona* Oct 2025 – Present

- C++, engine architecture, graphics, animation, AI and physics.

BA Digital Arts (Honours) — *University of the Witwatersrand* Jan 2021 – Dec 2024

SKILLS

Languages: C++17/20 (templates, memory management, OOP), HLSL, C#, TypeScript, SQL
Graphics: DirectX 12, Metal, deferred PBR, tiled light culling, shadow maps, compute shaders, post-processing
Engine: ECS, GPU resource lifetime, physics, A* pathfinding, hot-reload scripting, unit testing
Tools: Visual Studio, Xcode, PIX, RenderDoc, CMake, Git, GitHub Actions, Unity, Unreal (coursework)
Spoken languages: English (fluent), Spanish (beginner, currently learning)